nerve to say, ‘I think your Copenhagen interpretation is lousy,’” he recalled later. “Besides, the lift ride wasn’t very long. Now, if the lift had gotten stuck between floors, that would have made my day! In which way, I don’t know.”

When the Bells arrived for their sabbatical in California three years later, Bell took the opportunity away from his usual work at CERN to finally figure out where von Neumann had gone wrong, and to show up Jauch. He found that von Neumann’s revered proof, consistently invoked to defend against any heresy, was hardly a proof of anything at all. “The von Neumann proof, if you actually come to grips with it, falls apart in your hands!” said Bell. “There is *nothing* to it. It’s not just flawed, it’s *silly*!” The great John von Neumann, as it turned out, had simply made a mistake—he made assumptions in his proof that were entirely unwarranted. “When you translate [von Neumann’s assumptions] into terms of physical disposition, they’re nonsense.… The proof of von Neumann is not merely false but *foolish*!”

Bell didn’t merely show von Neumann and Jauch were wrong—he left a new proof in place of the old ones. Von Neumann’s proof and its ilk (including Jauch’s “strengthened” proof and the proof Jauch had mentioned to Bell, by Andrew Gleason) purported to rule out any interpretation of quantum physics that used so-called hidden variables. A hidden-variables interpretation assigns definite locations or other properties to quantum objects before they are observed, even if those properties can’t be calculated from the theory itself. These properties go unseen in the mathematics of quantum physics, hence “hidden” variables. Bohm’s pilot-wave interpretation is a prime example of such a theory: in Bohm’s world, particles always have positions, even though those positions are largely hidden from view and can’t be calculated from Schrödinger’s equation. The proofs of von Neumann, Jauch, and Gleason all suggested that this kind of scheme must be impossible—yet Bohm’s pilot-wave interpretation clearly worked, as Bell knew quite well. Something must be wrong, and Bell thought he knew what it was. He meticulously disassembled the no-hidden-